CCNA Day 4 Study Notes

IPv4 Addressing and Subnetting - Part 1

IPv4 addressing fundamentals - based on the TBN (TungaBadra Networks) Day 4 class handout - CCNA 200-301 Subnetting

1. IPv4 addressing basics

Each IPv4 address is **32 bits** long and is written in decimal format as **4 octets**. Each octet is 8 bits, so an octet can hold values from 0 to 255.

Bit values inside one octet

Bit position	7	6	5	4	3	2	1	0
Power	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Decimal value	128	64	32	16	8	4	2	1

2. Decimal to binary (and back)

Decimal to binary - turn bits on from the largest value down

Example 1: convert decimal 192 to binary.

- 192 > 128, so turn the 128 bit position on (1). Remainder: 192 - 128 = 64.
- 64 = 64, so turn the 64 bit position on (1). Remainder: 0, so all remaining bits are 0.
- Decimal 192 = binary 11000000.

Example 2: convert decimal 205 to binary.

- 205 > 128 -> 128 bit on. 205 - 128 = 77.
- 77 > 64 -> 64 bit on. 77 - 64 = 13.
- 13 < 32 and 13 < 16 -> 32 and 16 bits off.
- 13 > 8 -> 8 bit on. 13 - 8 = 5. 5 > 4 -> 4 bit on. 5 - 4 = 1.
- 1 < 2 -> 2 bit off. 1 = 1 -> 1 bit on.
- Decimal 205 = binary 11001101.

Binary to decimal - add the values of the ON bits

- 11101101 -> 128 + 64 + 32 + 8 + 4 + 1 = 237.
- 10001001 -> 128 + 8 + 1 = 137.

3. Subnet mask chart

A subnet mask marks which bits of the address are network bits (1s) and which are host bits (0s). Masks are always a run of 1s followed by a run of 0s, so each octet can only take these values:

Binary pattern	Decimal
00000000	0
10000000	128
11000000	192
11100000	224
11110000	240
11111000	248
11111100	252
11111110	254
11111111	255

Subnet masks and CIDR notation

Dotted decimal format	CIDR notation
255.0.0.0	/8
255.128.0.0	/9
255.192.0.0	/10
255.224.0.0	/11
255.240.0.0	/12
255.248.0.0	/13
255.252.0.0	/14
255.254.0.0	/15
255.255.0.0	/16
255.255.128.0	/17
255.255.192.0	/18
255.255.224.0	/19
255.255.240.0	/20
255.255.248.0	/21

Dotted decimal format	CIDR notation
255.255.252.0	/22
255.255.254.0	/23
255.255.255.0	/24
255.255.255.128	/25
255.255.255.192	/26
255.255.255.224	/27
255.255.255.240	/28
255.255.255.248	/29
255.255.255.252	/30

- CIDR masks are just prefix lengths - a /12 mask, for example, can be applied to any address. The prefix length does NOT decide the class of an address.
- Classful default masks (historical classful addressing): Class A /8 = 255.0.0.0, Class B /16 = 255.255.0.0, Class C /24 = 255.255.255.0. The class comes from the first bits of the address itself - first octet 1-126 = Class A, 128-191 = Class B, 192-223 = Class C (RFC 791).

4. Network address and broadcast address

Broadcast address

The broadcast address is the **last** address of an IP network. It addresses all the nodes at the same time - a packet sent to it is delivered to every node on the network. All bits of the host portion are set to one.

Network address

The network address uniquely identifies the network portion of an IP network and is the same for all hosts within the same network. All bits of the host portion are set to zero.

Walk-through: one Class C network

- Private Class C range: 192.168.0.0 - 192.168.255.255 (that is 192.168.0.0/16, i.e. 256 Class C networks of /24 each).
- First network with mask 255.255.255.0 (/24): network address 192.168.0.0; hosts 192.168.0.1 ... 192.168.0.254; broadcast 192.168.0.255.
- Total addresses: 256. Usable host addresses: 256 - 2 = 254.
- Next networks: 192.168.1.0 - 192.168.1.255, then 192.168.2.0 - 192.168.2.255, and so on up to the 256th network 192.168.255.0 - 192.168.255.255.

5. Finding the network address - the AND operation

A device finds the network address of any IP address by doing a binary **AND** between the IP address and its subnet mask. (1 AND 1 = 1; anything AND 0 = 0.)

Example 1: find the network address of host 192.168.10.50/24.

- IP: 11000000.10101000.00001010.00110010 (192.168.10.50)
- Mask: 11111111.11111111.11111111.00000000 (255.255.255.0)
- AND: 11000000.10101000.00001010.00000000 -> 192.168.10.0
- So 192.168.10.50/24 belongs to network 192.168.10.0/24.

Example 2: find the network address of host 10.205.79.90/22.

- IP: 00001010.11001101.01001111.01011010 (10.205.79.90)
- Mask: 11111111.11111111.11111100.00000000 (255.255.252.0)
- AND: 00001010.11001101.01001100.00000000 -> 10.205.76.0
- So 10.205.79.90/22 belongs to network 10.205.76.0/22.

6. Network, first host, last host and broadcast

- Network address: set all host bits to 0. Host bits are counted from right to left.
- Broadcast address: set all host bits to 1.
- First host: network address + 1 (turn on the right-most bit).
- Last host: broadcast address - 1.

Example 1: for host 192.168.199.100/24 find all four.

Item	Binary (host bits marked)	Decimal
IP address	11000000.10101000.11000110.01100100	192.168.199.100
Network	11000000.10101000.11000110.00000000	192.168.199.0
First host	11000000.10101000.11000110.00000001	192.168.199.1
Last host	11000000.10101000.11000110.11111110	192.168.199.254
Broadcast	11000000.10101000.11000110.11111111	192.168.199.255

Example 2: for host 102.224.96.32/20 find all four. (/20 means the network bits end inside the 3rd octet; mask 255.255.240.0.)

Item	Binary (3rd and 4th octets)	Decimal
IP address	01100000.00100000	102.224.96.32
Network	01100000.00000000	102.224.96.0
First host	01100000.00000001	102.224.96.1
Last host	01101111.11111110	102.224.111.254
Broadcast	01101111.11111111	102.224.111.255

7. Finding the next network - the adding factor

Rule: find the octet that holds the last network bits. The **adding factor** = 2^(number of host bits in that octet). Add the factor to that octet to step from one network to the next.

Example 1: next network after the network of 192.168.10.50/24.

- Mask 255.255.255.0 (/24): the 3rd octet holds the last network bits; host bits in that octet = 0, so adding factor = 2^0 = 1.
- Network address is 192.168.10.0/24. Next network: 10 + 1 = 11 -> 192.168.11.0/24.

Example 2: next network after the network of 190.168.10.50/21.

- Mask 255.255.248.0 (/21): the last network bits are in the 3rd octet, with 3 host bits there, so adding factor = 2^3 = 8.
- Network address: 10 falls inside the block 8 - 15, so the network is 190.168.8.0/21.
- Next network: 8 + 8 = 16 in the 3rd octet -> 190.168.16.0/21.

Class handout correction: in this example the handout writes the network address as '190.168.8.0/16' - the correct prefix is **/21** (the mask applied is 255.255.248.0). It also writes the next network as '190.169.16.0/21' - the second octet does not change; the correct answer is **190.168.16.0/21**.

8. Interview questions and answers

Q1. What is a subnet mask?

- A subnet mask identifies which bits of an IP address are network bits and which are host bits.
- Default masks: Class A 255.0.0.0, Class B 255.255.0.0, Class C 255.255.255.0.

Q2. What is CIDR notation and how is it used in IP addressing?

- CIDR (Classless Inter-Domain Routing) notation is a compact way to write an IP address with its network prefix length.
- It is written as the IP address, a slash, and the prefix length - for example 192.168.1.0/24.

Q3. What is the difference between the network address (ID) and the broadcast address (ID)?

- Network address: identifies the network portion itself; all host bits are 0. It cannot be assigned to a host.
- Broadcast address: addresses all nodes on that network at once; all host bits are 1. It cannot be assigned to a host either.

Q4. How does a system identify the network ID of an address? (added by Instinct - beyond the class handout)

- It performs a binary AND operation between the IP address and the subnet mask.
- Wherever the mask has 1s the IP bits are kept; wherever the mask has 0s the result is 0 - leaving exactly the network address.

Q5. Why are two addresses in every subnet not usable by hosts? (added by Instinct - beyond the class handout)

- The first address (all host bits 0) is the network address and the last (all host bits 1) is the broadcast address.
- That is why usable hosts = 2^(host bits) - 2.
- Scope note: the -2 rule covers normal host subnets (up to /30). Exceptions: a /31 point-to-point link has no network or broadcast address - both addresses are usable (RFC 3021) - and a /32 is a single host route.

Q6. What are the private IPv4 ranges? (added by Instinct - beyond the class handout)

- 10.0.0.0 - 10.255.255.255 (10.0.0.0/8 - one Class A-sized block).
- 172.16.0.0 - 172.31.255.255 (172.16.0.0/12 - 16 contiguous Class B-sized networks, not a single Class B network).
- 192.168.0.0 - 192.168.255.255 (192.168.0.0/16 - 256 Class C-sized networks).
- These ranges are defined by RFC 1918.

9. Quick revision sheet

- IPv4 = 32 bits = 4 octets of 8 bits. Bit values: 128 64 32 16 8 4 2 1.
- Mask octets can only be 0, 128, 192, 224, 240, 248, 252, 254, 255.
- Network address = IP AND mask. Broadcast = network with all host bits set to 1.
- First host = network + 1; last host = broadcast - 1; usable = 2^h - 2 (normal subnets up to /30; /31 point-to-point links and /32 host routes are the exceptions - RFC 3021).
- Adding factor for next network = 2^(host bits in the octet with the last network bit).
- Classful default masks: A /8, B /16, C /24. Private ranges (RFC 1918): 10/8, 172.16/12, 192.168/16.